GEOMETRIC GRAPH RESEARCH / PROOF CLOSURE

From outer triangles to deletion-resistant coverage

An explicit localization proof and a new robustness prediction

16 September 2026

Main advance. The earlier boundary-area argument left two interchanges unresolved: removing a spatial endpoint cutoff, and integrating a pointwise boundary limit over all query depths. This note supplies separate geometric bounds for both. It uses one-city deletion robustness rather than an unspecified polynomial bound on ray integrals.

Derived consequence. If $C_{n,2}$ is the intersection of all leave-one-city-out convex hulls and K_n the ordinary hull, the written argument gives

$$\frac{\mathbb{E}\,\mathrm{area}(D \setminus C_{n,2})}{\mathbb{E}\,\mathrm{area}(D \setminus K_n)} \longrightarrow 1.550394492\ldots$$

under the stated smoothness, positive-curvature, and positive-density assumptions. This is a ratio of expectations, not a deterministic ratio for each graph.

Independent numerical test. Predictions were frozen before 960 new iid configurations were generated. The robust core was computed by direct hull intersections; boundary triangles were constructed separately. The primary ratio approaches the prediction, while some absolute coefficients still miss their limiting values. Those misses remain in the report.

Proof status. Proofs below are supplied for independent review, not peer-reviewed or machine-verified results. The argument is restricted to a smooth, positively curved planar domain and an everywhere-positive continuous sampling density. It neither proves the entire intermediate fan process nor applies to polygonal corners. Priority and novelty have not been certified.

This archive contains only the new code, records, proof source, figures, validation and source map. Earlier simulation archives are not duplicated.

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1 The question this step resolves

The preceding work proposed a universal coefficient for the expected total area of faces sharing an edge with the random convex hull. Its coefficient was a convergent integral in a parabolic Poisson model. Independent simulations supported it, including nonuniform sampling. However, convergence of point clouds on bounded sets alone does not justify averaging a functional that may depend on distant cities or integrating over unbounded rescaled query depths.

This note makes the transfer explicit. The key observation is that membership in a hull-edge-adjacent face can be expressed through how many single-city deletions remove the query from the hull. That event is determined by halfplanes containing at most one city. Two occupied guard regions localize all such halfplanes; three occupied groups give a uniform tail bound in query depth.

Tukey/halfspace depth and its depth regions are established geometric-statistical concepts. They should not be confused with peeling all convex-hull vertices simultaneously, nor with a new statistic invented here. For background and distinctions between Tukey layers and convex layers, see [1]; the continuous depth-region formulation also appears in [2]. The particular area identity and boundary-functional proof below are stated and derived explicitly rather than inherited from those references.

2 Exact finite geometry

2.1 A core that survives any one-city deletion

Let $P = \{p_1, \dots, p_n\} \subset \mathbb{R}^2$, $n \geq 4$, be in general position. Write

$$K = \operatorname{conv}(P), \qquad K_{-i} = \operatorname{conv}(P \setminus \{p_i\}), \qquad C_2 = \bigcap_{i=1}^{n} K_{-i}.$$

A point of C_2 remains covered after *any one* city is removed. This is not simultaneous removal of all hull vertices. In the count convention for halfspace depth, C_2 is the depth-at-least-two region. Indeed, failure of membership in one K_{-i} is equivalent by strict separation to a halfplane containing the query and at most one original city. Closed-boundary conventions do not affect the area statements below.

For a query $q \in \operatorname{int} K$ off all city-pair lines, define

$$J(q) = \#\{i : q \notin K_{-i}\}.$$

Each such city is *individually essential*: deleting either of two essential cities alone removes the query. The terminology does not mean that two cities must be deleted together.

Lemma 1 (Angular characterization)**.** *For such a planar query and $n \geq 4$, $J(q) \leq 2$. Moreover, $J(q) = 2$ if and only if q lies inside the triangular face adjacent to a convex-hull edge of the complete finite-segment graph.*

Proof. Move q to the origin and sort the city directions. Let g_i be their successive circular gaps, summing to 2π. Since q is inside, every gap is below π. Deleting city i removes containment precisely when its two adjacent gaps join to exceed π.

Two disjoint adjacent-gap pairs cannot both exceed π, since their sum would exceed the total 2π. Among any three edges of a cycle of length at least four there are two disjoint edges. Hence at most two cities are essential, and when two are essential their directions must be adjacent.

Rotate the two adjacent directions to 0 and $\alpha \in (0, \pi)$. Their essentiality inequalities say that every other city direction lies strictly between π and $\pi + \alpha$. Equivalently, every remaining city z lies in the cone

$$Q_q(u, v) = \{z : q \in \operatorname{int} \operatorname{conv}(u, v, z)\}.$$

The line uv is supporting, and the two extremizing rays from its endpoints bound precisely the complete-graph face containing q. Conversely this cone condition makes both hull endpoints essential. The next construction verifies the face interpretation using finite segments. □

Place a hull edge at $u = (0, 0)$, $v = (L, 0)$, with every other city (x_i, y_i) satisfying $y_i > 0$. Set

$$a = \max_i x_i/y_i, \qquad b = \max_i (L - x_i)/y_i.$$

All remaining cities lie in $L - by \le x \le ay$. The supporting rays meet at

$$w = \left(\frac{aL}{a+b}, \frac{L}{a+b} \right).$$

Every other segment remains outside the open triangle uvw, and the two supporting segments reach its apex: their endpoint heights are at least $L/(a + b)$. Thus it is the adjacent face, of exact area

$$A_{uv} = \frac{L^2}{2(a+b)}. \tag{1}$$

For $n \ge 4$ different hull edges give different such faces. A face containing two consecutive hull edges would contradict one of these supporting-ray extrema in the presence of a fourth city.

2.2 Boundary triangles are exactly the double-counted deletion loss

Let B be total hull-edge-adjacent face area, $U = \operatorname{area}(K \setminus C_2)$, and

$$S_1 = \sum_i \operatorname{area}(K \setminus K_{-i}).$$

Writing $A_1 = \operatorname{area}\{q \in K : J(q) = 1\}$, Lemma 1 gives

$$S_1 = A_1 + 2B, \qquad U = A_1 + B.$$

Consequently

$$\boxed{B = S_1 - U = (n-1)\operatorname{area}(K) - \sum_i \operatorname{area}(K_{-i}) + \operatorname{area}(C_2).} \tag{2}$$

This identity holds for each generic finite drawing, without randomness or asymptotics. It is the area overcount caused by adding overlapping single-deletion losses.

2.3 The original seven-city drawing

Exact rational halfplane intersections, independent of the floating-point simulation, give

Region or sum	Exact area	Decimal area
Ordinary hull	38	38
Deletion-resistant core C_2	71351/4752	15.014941
Deletion-fragile region U	109225/4752	22.985059
Sum of individual losses S_1	61/2	30.5
Two-essential triangles B	35711/4752	7.514941
One-essential region A_1	36757/2376	15.470118

Thus 39.51% of the hull survives every single-city deletion. The remaining 60.49% is fragile to at least one deletion. Boundary triangles occupy 19.78% of the hull and are the portion lost under two different individual deletions. The displayed areas satisfy $30.5 - 22.9850589\ldots = 7.5149411\ldots$ exactly as rationals.

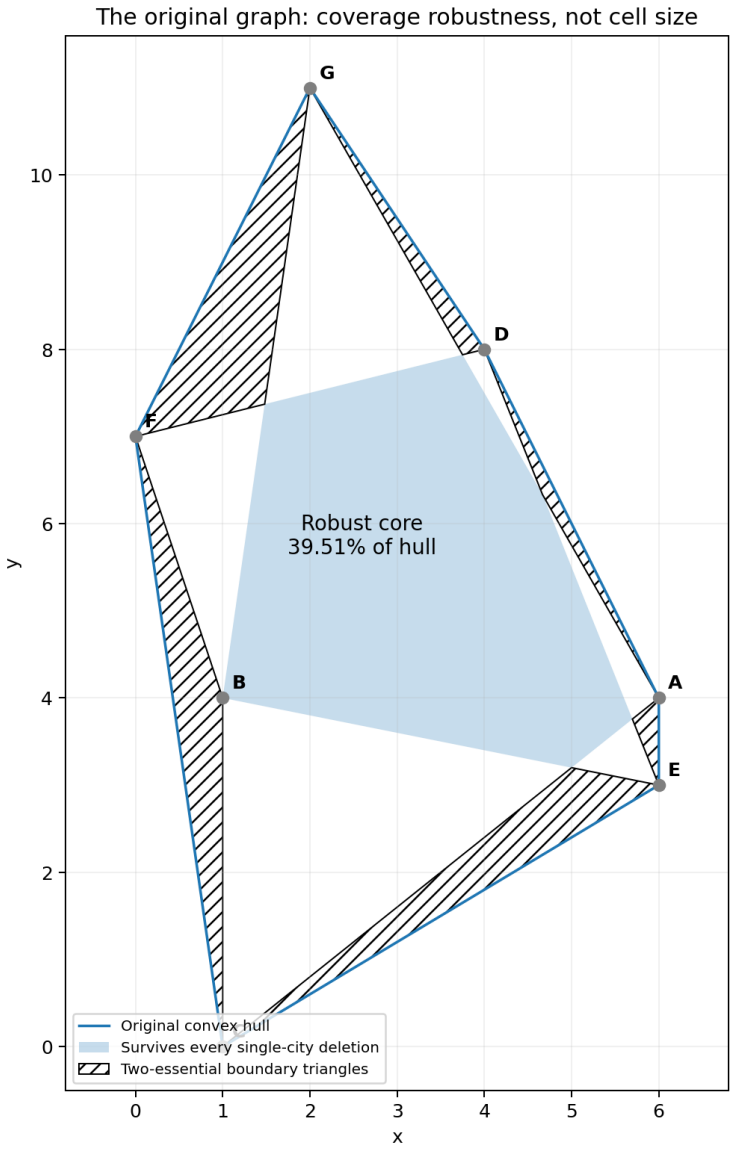


Figure 1: Original graph: the filled core survives every one-city deletion. Hatched boundary triangles form the two-essential region. Unfilled hull interior outside the core and triangles has one essential city.

3 Main boundary theorem and assumptions

Let $D \subset \mathbb{R}^2$ be a compact convex body with C^3 boundary and curvature bounded above and below by positive constants. Let $f : D \to (0, \infty)$ be continuous, integrate to one, and let $X_1, \ldots, X_n$ be iid with density f. Compactness makes f bounded above and away from zero. These conditions include a disk with uniform or linearly tilted positive density; they exclude corners, flat portions and densities vanishing at the boundary.

Define K_n, $C_{n,2}$ and B_n as above and set

$$M_n = \operatorname{area}(D \setminus K_n), \qquad M_{n,2} = \operatorname{area}(D \setminus C_{n,2}), \quad G(f,D) = \int_{\partial D} \kappa(s)^{1/3} f(s)^{-2/3}\, \mathrm{d}s.$$

Theorem 2 (Boundary area and deletion-resistant deficit)**.** *Under these assumptions,*

$$n^{2/3}\mathbb{E}B_n \longrightarrow c_B G(f,D), \tag{3}$$

$$n^{2/3}\mathbb{E}M_{n,2} \longrightarrow c_2 G(f,D), \tag{4}$$

where

$$c_M = (2/3)^{1/3}\Gamma(5/3), \qquad c_2 = \frac{5}{3}c_M - c_B,$$

and c_B is the explicit convergent integral in Section 7. Numerically,

$$c_M = 0.7886206526\ldots, \quad c_B = 0.0916946380\ldots, \quad c_2 = 1.2226731163\ldots.$$

Together with the classical weighted missing-area theorem,

$$\boxed{\frac{\mathbb{E}M_{n,2}}{\mathbb{E}M_n} \longrightarrow \frac{5}{3} - \frac{c_B}{c_M} = 1.5503944924\ldots.} \tag{5}$$

The weighted missing-area theorem is due to Böröczky–Fodor–Hug and is reproduced with its constant in Besau–Ludwig–Werner [4, Theorem 1.2]. It is used with geometric area as the target measure, not with sampling probability mass. The density factor is classical; the proof below checks the transfer for the separate complete-graph boundary and deletion-core functionals.

The status of Theorem 2 is a proof supplied here for review. The following sections state the uniform estimates explicitly; there is no appeal to the earlier simulations as a proof step.

4 First missing estimate: an integrable depth bound

Lemma 3 (Three groups protect a query)**.** *There exist $C, c, \delta_0 > 0$ such that for every query $q \in D$ at distance $\delta \le \delta_0$ from ∂D and all $n \ge 4$,*

$$\mathbb{P}(q \notin C_{n,2}) \le C(1 + n\delta^{3/2})e^{-cn\delta^{3/2}}. \tag{6}$$

For queries at distance at least δ_0, the bound can be replaced by $C(1+n)e^{-cn}$.

Proof. At the nearest boundary point p, use tangent coordinate u and inward coordinate z. Put $u = \sqrt{\delta/\kappa(p)}\, x$ and $z = \delta y$. The query is $(0,1)$; uniformly in p, the boundary converges on bounded coordinate sets to $y = x^2/2$ as $\delta \downarrow 0$.

Consider the three closed rectangles

$$R_- = [-.55, -.45] \times [.35, .45],$$
$$R_+ = [.45, .55] \times [.35, .45],$$
$$R_0 = [-.05, .05] \times [1.75, 1.85].$$

They lie strictly above the parabola, with minimum clearance 0.19875. Every triangle with one point in each rectangle contains $(0,1)$ strictly. To check the latter directly, the top joining edge lies below $y = .45$. The other two edges cross $y = 1$ strictly on opposite sides of the vertical axis: their extreme choices are

bounded away from $x = 0$. Equivalently, the relevant oriented determinants are multilinear in the endpoint coordinates, so it is enough to check their signs at the 64 choices of corners. The minimum point-to-edge distance for those extreme triangles is over 0.205; this is recorded as an independent arithmetic check, not an assumption of the lemma.

Uniform Taylor control supplies a common δ_0 for which the corresponding physical rectangles lie inside D. Each has area $0.01\,\delta^{3/2}/\sqrt{\kappa(p)}$. Therefore its sampling probability p_j lies between $a\delta^{3/2}$ and $b\delta^{3/2}$ for uniform constants $0 < a \le b$.

If each rectangle contains at least two cities, every single deletion leaves at least one city in each, and their triangle still contains q. Consequently $q \in C_{n,2}$. No independence between rectangle counts is needed. A union bound and the binomial marginal law give

$$\mathbb{P}(q \notin C_{n,2}) \le \sum_{j=1}^{3} \left[(1-p_j)^n + np_j(1-p_j)^{n-1}\right] \le 3(1 + ebn\delta^{3/2})e^{-an\delta^{3/2}}.$$

This is (6) after renaming constants. For a query in the fixed interior, three small disks at the vertices of an equilateral triangle centered on the query provide the same enclosure. A uniform disk radius is available from the distance δ_0 to the boundary, and f is bounded below. Their binomial failure probabilities give the stated exponential bound. □

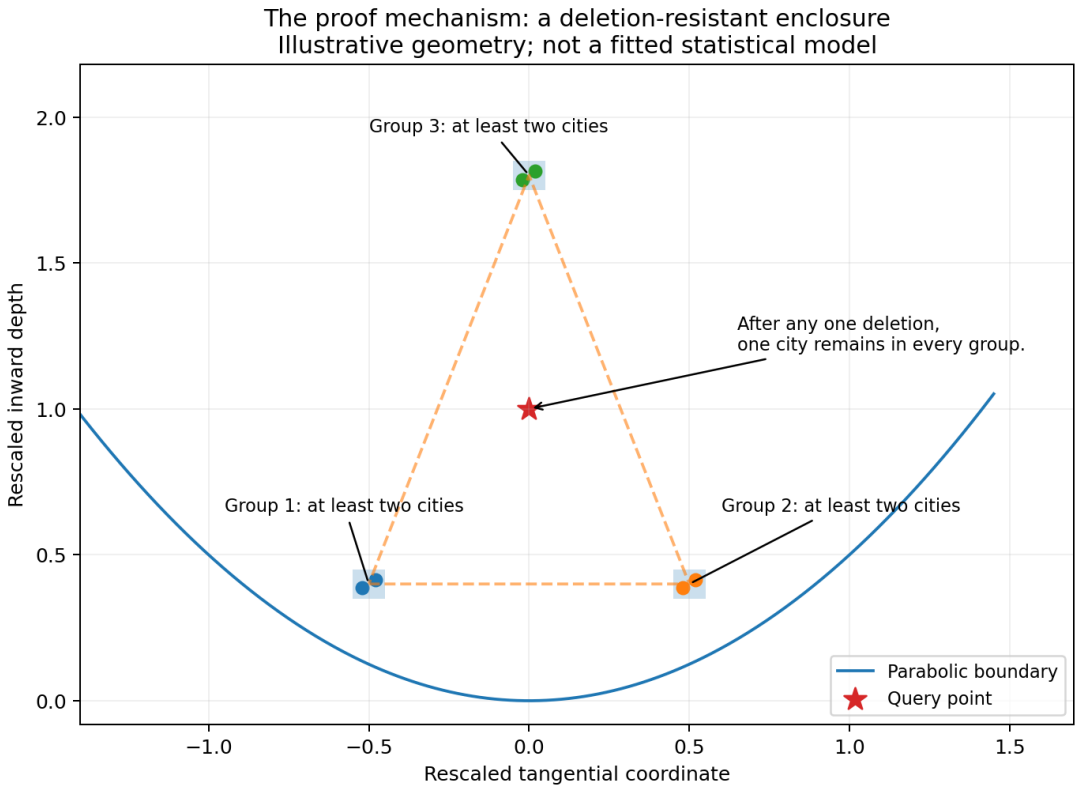


Figure 2: The deterministic enclosure behind the probability estimate. Counts are allowed to be multinomially dependent. One deletion cannot empty a group initially containing two cities.

The event that the query lies in a boundary triangle is contained in $\{q \notin C_{n,2}\}$. The same bound therefore controls boundary membership, ordinary hull failure, and the expected number of essential cities (the last with a factor of at most two when inside).

5 Second missing estimate: distant endpoints cannot decide the local event

At a fixed boundary point p, write

$$a_n = (nf(p)\kappa(p))^{-1/3}, \qquad b_n = \kappa(p)^{1/3}(nf(p))^{-2/3}.$$

Thus $nf(p)a_nb_n = 1$ and $\kappa(p)a_n^2 = b_n$. In coordinates $(u, z) = (a_nx, b_ny)$, the binomial point cloud converges on bounded sets to a unit-intensity Poisson process Π on

$$\mathcal{P} = \{(x, y) : y \geq x^2/2\}.$$

For disjoint bounded sets, this follows directly by the multinomial rare-event limit and uniform convergence of the transformed density. The parabolic scale and broader framework are classical [3].

Compact-set convergence is not by itself sufficient for deletion events. The next lemma supplies the missing stabilization.

Lemma 4 (Two occupied guards localize low-count halfplanes)**.** *Fix $t > 0$ and the transformed query $(0, t)$. For arbitrarily large L, the events "outside the hull" and "inside with $J = j$", $j = 0, 1, 2$, are determined by the points in a deterministic box of width $O(L)$ and height $O(L^2)$, except on an event whose limiting upper probability is*

$$C(1 + L^3)e^{-cL^3}.$$

The constants may be chosen uniformly in the boundary point. No whole-chord-process local finiteness is asserted.

Proof. Choose L with $t \leq L^2/4$. Use the guard rectangles

$$G_+ = [L, 1.1L] \times [L^2, 1.1L^2], \qquad G_- = [-1.1L, -L] \times [L^2, 1.1L^2].$$

They lie strictly inside $\mathcal{P}$ and have area $0.01L^3$. They also lie in the transformed finite domain for all sufficiently large n (for fixed L). The probability either guard contains fewer than two points is at most $C(1 + L^3)e^{-cL^3}$ in the limit. This follows from Poisson probabilities, or the corresponding binomial count bound. Call the complementary event G.

On G, any halfplane through $(0, t)$ containing at most one point must be an *outward* cap $y < t + sx$. An upward halfplane $y > t + sx$ contains an entire guard: the left one for $s \geq 0$, the right one for $s < 0$. A vertical halfplane also contains one guard. For the outward cap, $s \geq 2L$ includes G_+ and $s \leq -2L$ includes G_-. Thus every potentially relevant cap has $|s| < 2L$.

There is a uniform constant $c_0 > 0$ for which the entire transformed domain obeys $y \geq c_0x^2$. To see this, in physical coordinates the strictly supporting tangent meets the compact strictly convex domain only at p. Taylor expansion gives $z \geq cu^2$ near p uniformly in p; compactness gives the same inequality, with a possibly smaller $c > 0$, away from p. Scaling divides c by the bounded curvature $\kappa(p)$.

Every relevant cap therefore lies in

$$c_0x^2 \leq y < t + 2L|x|.$$

Solving the quadratic inequality bounds $|x| \leq C_0L$ and then $0 \leq y \leq C_1L^2$. This is the promised deterministic box.

A zero-point halfplane detects failure of ordinary hull containment. A one-point halfplane, in the absence of a zero-point one, identifies an essential city. Every separating halfplane can be moved until its boundary passes through the query without increasing its city count. Thus these events are identical for the full point cloud and its restriction to the box on G. The guards belong to that restriction, so the same argument applies in both directions. □

Corollary 5 (Pointwise boundary classification limit)**.** *At $q_n = p + tb_n\nu_{\mathrm{in}}(p)$, probabilities of the four states (outside, $J = 0$, $J = 1$, $J = 2$) converge to their unit-intensity parabolic Poisson counterparts.*

Proof. First fix L. In the guard box, point-process convergence applies. The finite-point classification is continuous except when a query lies on a line through two sites, a site lies on a box boundary, or related exact degeneracies occur. Each has limiting probability zero. The error in using the box is at most the bound in Lemma 4, both in the finite model in limiting upper probability and in the Poisson model. Send $n \to \infty$, then $L \to \infty$. □

This localizes *the low-count halfplane event*, not every segment of the graph. It bypasses the unresolved endpoint-ray integration tails in the earlier proof attempt.

6 Integrating the local limit: the missing interchange is now controlled

Let $p_2(t)$ denote the parabolic probability of state $J = 2$ and let $p_{<2}(t)$ denote failure of the parabolic deletion-resistant core.

In a fixed boundary tube, use normal coordinates (p, δ) with area Jacobian $1 - \kappa(p)\delta$. At each p set $\delta = b_n(p)t$. Then

$$n^{2/3}b_n(p) = \kappa(p)^{1/3}f(p)^{-2/3}.$$

By Lemma 3, both the finite boundary-face probability and the core-failure probability are bounded after this change of variables by

$$C(1 + t^{3/2})e^{-ct^{3/2}},$$

with constants uniform in p, n. This function is integrable on $[0, \infty)$. The Jacobian is uniformly bounded and tends to one. Outside the tube, the interior bound is exponentially small even after multiplication by $n^{2/3}$.

Corollary 5 and dominated convergence now give

$$\lim_n n^{2/3}\mathbb{E}B_n = G(f, D)\int_0^\infty p_2(t)\,\mathrm{d}t,$$
$$\lim_n n^{2/3}\mathbb{E}M_{n,2} = G(f, D)\int_0^\infty p_{<2}(t)\,\mathrm{d}t.$$

This proves the existence and the geometric factor in both limits. The next two sections evaluate the two universal coefficients. In particular, the integration is not justified merely by saying the limiting ray integral converges; it is justified by a uniform bound on the finite-city probabilities.

7 The boundary coefficient is the same integral as before

For the query $(0, t)$ in the parabolic process, a qualifying hull pair must straddle its vertical line; otherwise the opposite cone omits an infinite-intensity portion of $\mathcal{P}$. Parameterize its two rays by slopes $a > b$. Set

$$R_\pm(s, t) = \sqrt{s^2 + 2t} \pm s, \qquad J_+(s, t) = \frac{R_+(s, t)^2(2\sqrt{s^2 + 2t} - s)}{6}.$$

Here J_+ is the right cap area under $y = t + sx$ and above the parabola. The required cone is $y \geq \max(t+ax, t+bx)$. Integrating the candidate endpoints along its two opposite rays contributes $R_-(a, t)^2/2$

and $R_+(b,t)^2/2$. The two-point Mecke identity [5, Chapter 4], together with Lemma 1, counts the unique qualifying hull pair and yields

$$p_2(t) = \frac{1}{4}\int_{a>b} R_-(a,t)^2 R_+(b,t)^2 e^{-J_+(a,t)-J_+(-b,t)}\,\mathrm{d}a\,\mathrm{d}b.$$

This is a point-process calculation for sites, not an assumption that crossings are independent.

Put $a=\sqrt{2t}\sinh\alpha$, $b=\sqrt{2t}\sinh\beta$, $m=(\alpha+\beta)/2$, $d=(\alpha-\beta)/2>0$, and

$$Q(m,d)=\frac{e^{3d}\cosh(3m)+3e^{d}\cosh(m)}{6}.$$

The t integration is an elementary Gamma integral. Thus

$$\boxed{c_B=\frac{\Gamma(8/3)}{12}\int_0^\infty\int_{-\infty}^\infty \frac{e^{-4d}[\cosh(2m)+\cosh(2d)]}{Q(m,d)^{8/3}}\,\mathrm{d}m\,\mathrm{d}d.} \tag{7}$$

For an explicit truncation bound, $Q\ge e^{3d+3|m|}/12$ implies that the integrand is at most

$$12^{8/3}\big(e^{-12d-6|m|}+e^{-10d-8|m|}\big).$$

Omitting $|m|>L$ or $d>D$ therefore costs no more than

$$\frac{\Gamma(8/3)}{12}\,12^{8/3}\left(\frac{e^{-6L}}{36}+\frac{e^{-8L}}{40}+\frac{e^{-12D}}{36}+\frac{e^{-10D}}{40}\right).$$

At $L=D=8$, this is below 3.75×10^{-21}. Quadrature discretization is checked separately by increasing orders and cutoffs; it is not covered by that analytic truncation bound. The orders 180, 260 and 320 reproduce the printed coefficient to the displayed accuracy. No coefficient is fitted to the production experiment.

8 A new consequence: the cost of one-deletion robustness

Let $p_0(t)$ be the parabolic probability of being outside the ordinary hull and let $j(t)$ be the expected number of individually essential points, set to zero when outside. From the finite angular argument and localization,

$$p_{<2}(t)=p_0(t)+j(t)-p_2(t).$$

To integrate j, vary the parabolic Poisson intensity λ. The map

$$(x,y)\mapsto(\lambda^{1/3}x,\lambda^{2/3}y)$$

preserves $\mathcal{P}$ and converts intensity λ into intensity one. Thus

$$p_{0,\lambda}(t)=p_0(\lambda^{2/3}t).$$

The Poisson perturbation/Mecke identity [5, Chapters 4 and 19] gives

$$\lambda\frac{\partial}{\partial\lambda}\mathbb{P}_\lambda(q\in\operatorname{conv}\Pi)=\mathbb{E}_\lambda[J(q)\,1_{\{q\in\operatorname{conv}\Pi\}}].$$

Here is one way to justify its use on the unbounded parabolic domain. Apply the finite-measure identity first on increasing bounded boxes. Integrate it over an intensity interval bounded away from zero. A planar

query has at most three present pivotal points for hull containment in any finite box by Carathéodory's theorem. The guard localization makes the classification stabilize almost surely and its expectation converge. Dominated convergence in the intensity integral then supplies the derivative identity for almost every intensity, which is sufficient below. Continuity, or the explicit smooth hull-gap density, yields the pointwise version.

At $\lambda = 1$,

$$j(t) = -\frac{2}{3} t\, p_0'(t).$$

The three-patch tail bound makes integration by parts legitimate; $tp_0(t)$ vanishes at both endpoints. Since $\int_0^\infty p_0(t)\,\mathrm{d}t = c_M$ (equivalently the classical missing-area coefficient under this normalization),

$$\int_0^\infty j(t)\,\mathrm{d}t = \frac{2}{3} c_M.$$

Consequently

$$\int_0^\infty p_{<2}(t)\,\mathrm{d}t = c_M + \frac{2}{3} c_M - c_B = \frac{5}{3} c_M - c_B.$$

This completes the proof of Theorem 2 as written.

A pitfall avoided. The exact finite identity also gives

$$\mathbb{E}B_n = n(\mathbb{E}M_{n-1} - \mathbb{E}M_n) + \mathbb{E}M_n - \mathbb{E}M_{n,2}.$$

It would not be valid to differentiate only the leading asymptotic $\mathbb{E}M_n \sim c_M G n^{-2/3}$ without controlling its remainder. The argument above instead proves the integrated pivotal coefficient directly inside the localized Poisson limit.

8.1 Other coefficients, with the same geometric factor

The calculation gives

Functional	Coefficient multiplying $G(f,D)n^{-2/3}$
Ordinary missing area	$c_M = 0.7886206526$
Sum of one-city deletion losses	$\frac{2}{3}c_M = 0.5257471017$
One-essential area	$\frac{2}{3}c_M - 2c_B = 0.3423578258$
Two-essential boundary triangles	$c_B = 0.0916946380$
Inside-hull deletion-fragile region	$\frac{2}{3}c_M - c_B = 0.4340524637$
Total missing from deletion-resistant core	$\frac{5}{3}c_M - c_B = 1.2226731163$

Thus about 21.125% of the *mean fragile-region area* is double-essential asymptotically:

$$\frac{\mathbb{E}B_n}{\mathbb{E}\,\mathrm{area}(K_n \setminus C_{n,2})} \longrightarrow \frac{c_B}{(2/3)c_M - c_B} = 0.2112524306\ldots.$$

The primary ratio $1.550394\ldots$ means about 55% *more missing area*, not loss of 55% of the full domain. Both missing areas tend to zero.

8.2 A limited extension

For a fixed integer k, the intersection of all hulls obtained by deleting $k-1$ cities is the depth-at-least-k region. Requiring at least k sites in every enclosure or guard rectangle repeats the localization argument. The binomial failure bound acquires a polynomial of degree $k-1$ in $n\delta^{3/2}$ while retaining its exponential tail. This establishes the same $n^{-2/3}$ rate with a finite parabolic-profile constant for each fixed k. No new explicit constant for $k \geq 3$, growing-k theorem, or priority claim is made here.

9 One frozen independent numerical experiment

9.1 Design and implementation

The primary prediction $1.5503944924\ldots$ and secondary prediction $0.2112524306\ldots$ were computed and saved before production. The experiment uses $n = 128, 512, 2048, 8192$, two disk densities $f_\eta(x, y) = (1 + \eta x)/\pi$ with $\eta = 0, 0.8$, and 120 independent graphs per group. The tilted density has a 9:1 contrast between its most and least populated boundary points. Exact rejection sampling is used with no minimum point-separation condition.

Every $C_{n,2}$ is computed directly by intersecting the leave-one-out hulls. Only deletion of an original hull vertex can change K_n, so the other deletions need no computation. Boundary triangles are independently computed by the support-ray formula (1). The identity (2) is checked, not used to manufacture the core area. All 960 configurations are retained. They contain 36,307 boundary triangles; their maximum area-identity discrepancy is 2.37×10^{-14}.

Ratios are ratios of sample means. Confidence intervals use 5,000 whole-graph bootstrap resamples with a fixed seed, pairing numerator and denominator. They are pointwise Monte Carlo intervals, not simultaneous bands, asymptotic-bias bounds, or probabilities that a theorem is true. Counts were not increased after inspection.

Density η	n	Graphs	Mean deficit ratio $M_{n,2}/M_n$	95% interval
0	128	120	1.530706	[1.504749, 1.556616]
0	512	120	1.538075	[1.518385, 1.558110]
0	2048	120	1.536716	[1.519439, 1.554867]
0	8192	120	1.544990	[1.531676, 1.558649]
0.8	128	120	1.496714	[1.474083, 1.520888]
0.8	512	120	1.526537	[1.502742, 1.552246]
0.8	2048	120	1.548713	[1.527511, 1.571132]
0.8	8192	120	1.548250	[1.532666, 1.565352]

In the table heading the ratio denotes a ratio of means, not the expectation of per-graph ratios. The tilted $n = 128$ interval excludes the limiting prediction. At $n = 8192$, both primary intervals include it.

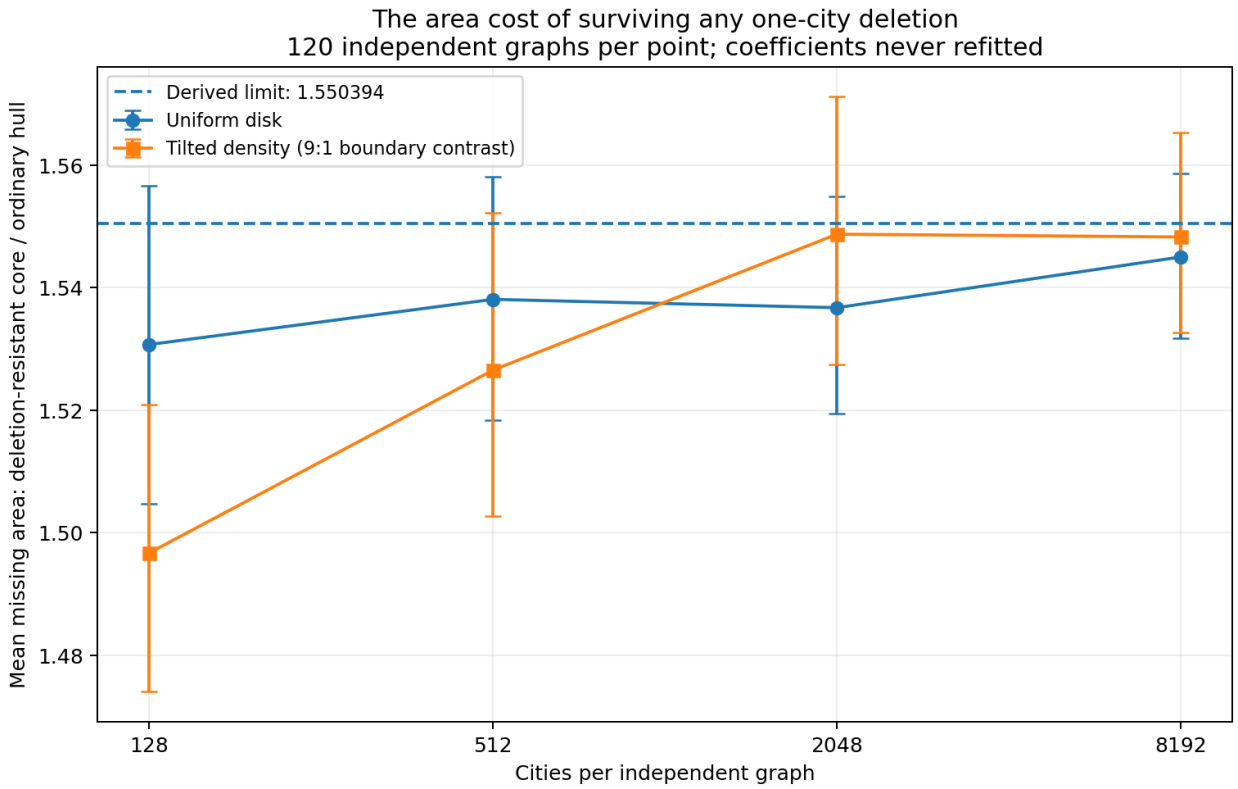


Figure 3: The primary frozen prediction. A visible small-n departure is retained rather than removed by fitting.

9.2 An absolute-coefficient check still misses

Agreement of the ratio does not guarantee that both absolute means already have their limiting coefficients. At $n = 8192$, the uniform-disk estimate of

$$n^{2/3}\mathbb{E}M_{n,2}/G(f, D)$$

is 1.198028, with interval $[1.183912, 1.211730]$, below $c_2 = 1.222673$ and outside that interval. The tilted estimate is 1.209770, with interval $[1.189774, 1.229768]$.

No exact finite-n error rate is proved here, and this fixed experiment does not isolate remaining asymptotic bias from sampling fluctuation. Thus the numerical evidence supports the primary ratio, but should not be described as every normalization matching. A finite-size miss does not refute an asymptotic limit without a corresponding rate bound; nor may it be hidden.

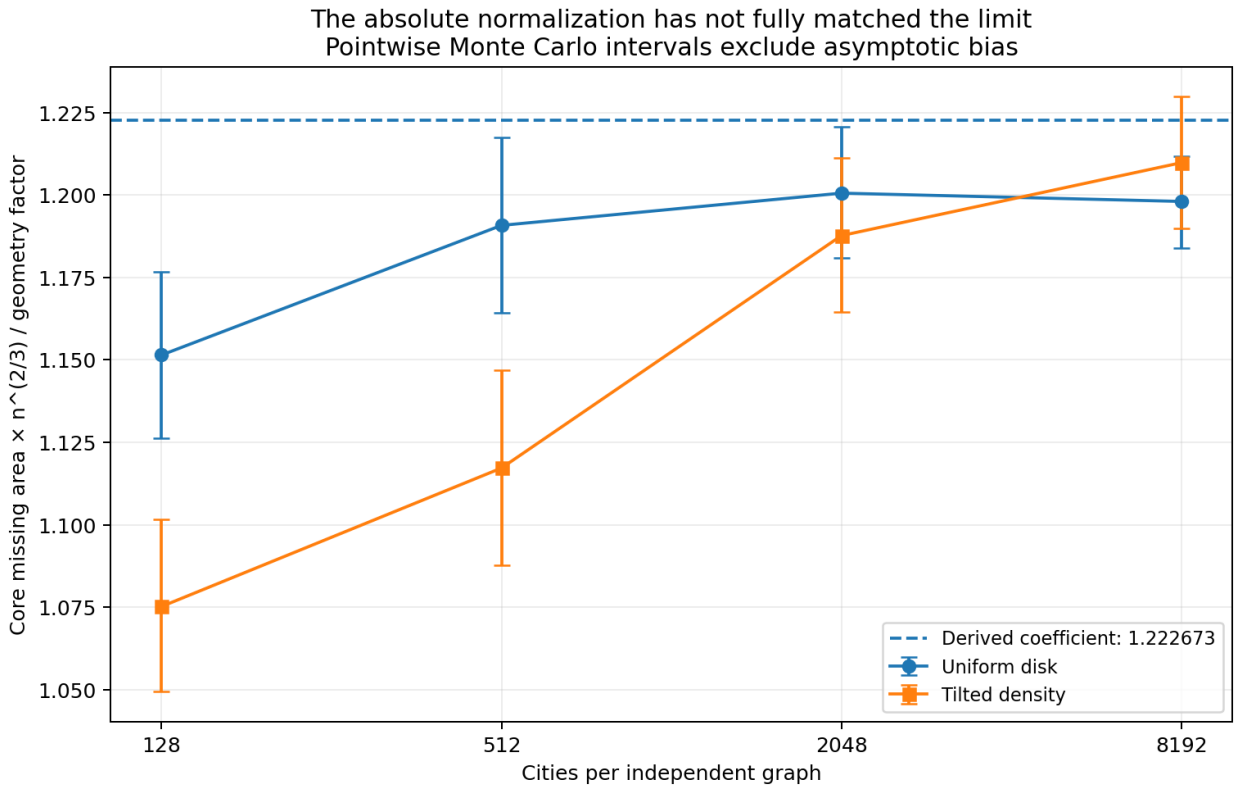


Figure 4: The absolute normalization is a harder diagnostic. In particular the uniform-disk interval at the largest tested size remains below the predicted limit.

9.3 Independent finite-geometry validation

A separate sample of 180 clouds used disk, tilted-disk and Gaussian distributions and sizes 4, 7, 10 and 13. Full complete-graph segment noding and polygonization confirmed all 952 support-constructed boundary triangles. The worst relative symmetric-difference discrepancy was 1.59×10^{-13}. For 3,600 independent query checks, angular deletion classification agreed with geometric membership in both the robust core and the boundary triangles, with no mismatches. These are implementation checks; the deterministic proofs do not rely on observing zero failures.

10 Literature, scope and next review obligations

The depth-region interpretation is established; it is not new merely because it emerged from our graph drawing. Published work on Tukey layers [1] studies related objects, but layers of sample points, convex peeling, and the continuous deletion core should not be interchanged. Random-hull parabolic scaling and localization methods are classical [3]. The weighted missing-area factor is explicitly available from [4]. Poisson perturbation and Mecke identities are standard [5].

The specific new work presented for review is the event-level proof linking the complete-graph boundary functional to the parabolic coefficient: three-group depth domination, two-guard endpoint localization, and the resulting depth-two deficit calculation. The public-web searches and primary sources examined did not certify whether the numerical depth-two coefficient or these exact formulations have appeared elsewhere. No “first discovery” claim follows from that search.

The prior angular-integration experiment is not represented as rerun here. Its finite-n formula is compatible with this argument, but numerical trapezoidal evaluations must not be called exact numerical probabilities without discretization control. The present proof and production test do not depend on those earlier approximate angular values.

This note does not close the full intermediate fan-process proof, establish a Brownian-frontier connection, settle boundary corners or zero curvature, determine a largest-face distribution, give a sharp finite-n rate, or prove a CLT for the new boundary functional. These are separate questions.

The most useful next review is adversarial examination of the two geometric lemmas and the Poisson pivotal-derivative transfer, followed by a specialist priority check on low empirical Tukey-depth regions. More regression sweeps would not resolve those issues.

11 Reproducibility and archive contents

The archive supplies the frozen plan, all 960 graph-level records and deterministic seed rules, exact rational original-graph calculations, theory quadrature and convergence checks, validation code and outputs, summary code, four figures, this source and compiled report, source map, environment versions and file hashes. It excludes earlier archives, downloaded third-party texts, font files and temporary LaTeX outputs. Run the scripts in the order listed in the README. A reproduction generates actual sampled city clouds, their ordinary and deletion hulls, and the boundary triangles; it does not construct all interior cells at large n.

References

[1] Z. Guo, Y. Li and S. Pei. Expected size of random Tukey layers and convex layers. *Computational Geometry* 103 (2022), 101856. `https://arxiv.org/abs/2008.02258`. Background on halfspace-depth conventions and the distinction from convex layers; not an asserted source for the new coefficient.

[2] Y. Gao and O. Sheffet. Private Approximations of a Convex Hull in Low Dimensions. Preprint (2020). `https://arxiv.org/abs/2007.08110`. Uses continuous Tukey-depth regions; not the boundary asymptotic proved here.

[3] P. Calka, T. Schreiber and J. E. Yukich. Brownian limits, local limits and variance asymptotics for convex hulls in the ball. *Annals of Probability* 41 (2013), 50–108. `https://arxiv.org/abs/0912.4339`. Parabolic random-hull framework and stabilization; graph-specific deletion classification still needs checking.

[4] F. Besau, M. Ludwig and E. M. Werner. Weighted floating bodies and polytopal approximation. *Transactions of the American Mathematical Society* 370 (2018), 7129–7148. Theorem 1.2, attributed there to Böröczky–Fodor–Hug. `https://arxiv.org/abs/1611.04132`. Classical weighted missing-area theorem and coefficient.

[5] G. Last and M. Penrose. *Lectures on the Poisson Process*. Cambridge University Press (2017), Chapters 4 and 19. `https://stoch.math.kit.edu/img/Last/lastpenrose2017.pdf`. Mecke identities and intensity perturbation. The bounded-box-to-parabolic transfer is supplied in this note.